

Amortization collapse for reference stabilized geometric Rényi superchannel divergences

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October 9, 2026

DOI: [10.5281/zenodo.23215121](https://doi.org/10.5281/zenodo.23215121)

Abstract

For every real order $1 < \alpha \leq 2$, we prove that the reference stabilized geometric Rényi divergence of two finite dimensional deterministic physical superchannels equals its nested amortized extension. Joint inserted channels $AR \rightarrow BR$ and the external state references in the inner channel divergences range over all positive finite dimensions. The proof combines a supported matrix perspective transformer inequality with a comb contraction, a completion of positive testers, and a physical realization of positive definite testers. It includes singular supported operators and gives value $+\infty$ for unsupported comb pairs. In the supported case, the ordinary supremum already has the same value at reference dimension $|C||A||B|$. A conventional proof is presented alongside a correspondence with a Lean 4 theorem and a reproducible source artifact. The result concerns the nested quantity of Hirche’s Eq. (39); it does not establish collapse of the larger fully amortized quantity in Eq. (40).

1 Introduction and scope

Amortization measures the distinguishability gained by a transformation after subtracting distinguishability already present in its inputs. For geometric Rényi divergence, Fang and Fawzi established a Choi formula, a channel chain rule, and the resulting collapse of ordinary channel amortization [2, Lemmas 5–7]. For superchannels, both a channel and a state can carry input distinguishability. Hirche distinguishes several corresponding amortized quantities [3, Definitions 4.4–4.5].

We establish equality between the reference stabilized ordinary quantity in Hirche’s Eq. (33) and the nested amortized quantity in Eq. (39). The additional argument beyond ordinary channel collapse is a uniform bound on a tester obtained by contracting the insertion and state perspectives. Its marginal slack can be completed to a normalized feasible tester. Conversely, a positive definite feasible tester can be realized with a single common input and output reference, using a pure initial state and a globally trace preserving joint insertion. These two steps turn the comb moment bound into the required equality of operational suprema.

*The research and proofs were developed jointly by GPT-6.1-Sol and Claude Opus 5.5 agents in an automated research run initiated by Vinícius Mohr. GPT-6.1-Sol wrote this manuscript. See the contribution and AI disclosure statement.

The QIQCOP record `op_1482756b02794495` cites Hirche’s question and suppresses the inserted-reference labels in its superchannel formulas [4]. We state the reference convention explicitly. Restricting the insertion to $A \rightarrow B$ changes the problem: an external reference attached only to the output channel’s state tests does not supply a reference through the insertion. The theorem below resolves the intended reference stabilized nested question. It must not be read as an assertion for that restricted convention. Nor does it address the fully amortized quantity in Hirche’s Eq. (40), arbitrary multi-slot combs, or the endpoint $\alpha = 1$.

The operator transformer estimate and channel chain-rule machinery are established ingredients [2, Lemma 47 and Lemmas 5–7]. We give their supported singular versions in the form used by the formal proof. Our contribution here is the superchannel tester argument and its complete formal verification. A bounded search of primary sources did not locate an earlier proof of this precise equality; this is not an exhaustive priority claim. The accompanying preparation record describes that search and its limits.

2 Definitions and the main theorem

All spaces are complex Hilbert spaces of positive finite dimension. All logarithms are to base two. A channel is a complex-linear completely positive trace preserving map. A state is positive semidefinite with trace one. Write $X \ll Y$ when $\text{supp } X \subseteq \text{supp } Y$ for positive semidefinite operators. The symbol Y^+ denotes the Moore–Penrose inverse, and $Y^{+1/2}$ its positive square root, zero on the kernel of Y .

Fix $1 < \alpha \leq 2$ and put $\kappa = \alpha - 1$. For positive semidefinite X, Y , define the matrix perspective and its moment by

$$G_\alpha(X, Y) = Y^{1/2}(Y^{+1/2}XY^{+1/2})^\alpha Y^{1/2}, \quad M_\alpha(X, Y) = \text{Tr } G_\alpha(X, Y). \quad (1)$$

On supported states the geometric divergence is

$$\widehat{D}_\alpha(\rho \parallel \sigma) = \frac{1}{\kappa} \log_2 M_\alpha(\rho, \sigma); \quad \widehat{D}_\alpha(\rho \parallel \sigma) = +\infty \quad \text{if } \rho \not\ll \sigma. \quad (2)$$

Although the matrix expression in (1) is defined for all positive semidefinite pairs, only supported pairs are assigned a finite divergence by (2).

For channels $N, M : P \rightarrow Q$, let

$$d_\alpha(N \parallel M) = \sup_{S, \rho_{SP}} \widehat{D}_\alpha((\text{id}_S \otimes N)(\rho) \parallel (\text{id}_S \otimes M)(\rho)), \quad (3)$$

$$d_\alpha^A(N \parallel M) = \sup_{\substack{S, \rho_{SP}, \sigma_{SP} \\ \widehat{D}_\alpha(\rho \parallel \sigma) < \infty}} \left\{ \widehat{D}_\alpha((\text{id}_S \otimes N)(\rho) \parallel (\text{id}_S \otimes M)(\sigma)) - \widehat{D}_\alpha(\rho \parallel \sigma) \right\}. \quad (4)$$

Here and below S ranges over every positive finite dimension. The finite-cost restriction permits every finite real cost and retains negative differences. In particular it never subtracts ∞ from ∞ .

A physical deterministic superchannel taking $A \rightarrow B$ channels to $C \rightarrow D$ channels has a realization

$$\Theta(N) = Q_\Theta \circ (N \otimes \text{id}_{E_\Theta}) \circ P_\Theta, \quad P_\Theta : C \rightarrow AE_\Theta, \quad Q_\Theta : BE_\Theta \rightarrow D, \quad (5)$$

where both maps are channels and the inaccessible memory E_Θ is finite. Every finite dimensional deterministic physical supermap has such a realization by the established realization theorem [1,

Theorem 1, Eq. (23)]. This representation theorem is cited as background; it is not a new formal theorem of our artifact. Each superchannel in a pair may have its own memory and its own realization.

For $N : AR \rightarrow BR$, the extended action $\Theta^R(N) : CR \rightarrow DR$ applies $P_\Theta \otimes \text{id}_R$, moves R past E_Θ , applies $N \otimes \text{id}_{E_\Theta}$, and applies $Q_\Theta \otimes \text{id}_R$ after reversing the corresponding reordering. The insertion acts on the entire AR input; it need not have product form. Define

$$U_\alpha(\Theta_1 \parallel \Theta_2) = \sup_{R, N: AR \rightarrow BR, \rho_{CR}} \widehat{D}_\alpha(\Theta_1^R(N)(\rho) \parallel \Theta_2^R(N)(\rho)), \quad (6)$$

$$\mathcal{A}_\alpha(\Theta_1 \parallel \Theta_2) = \sup_{\substack{R, N, M: AR \rightarrow BR \\ d_\alpha^A(N \parallel M) \in \mathbb{R}}} \left\{ d_\alpha^A(\Theta_1^R(N) \parallel \Theta_2^R(M)) - d_\alpha^A(N \parallel M) \right\}. \quad (7)$$

The inserted reference R ranges over all positive finite dimensions. Each occurrence of d_α^A independently optimizes its external state reference S . The subtracted term in (7) is the entire input-channel supremum, not the cost of a selected state test. These are precisely the domains of Hirche's Eqs. (33) and (39) [3].

Theorem 1 (Reference stabilized nested collapse). *For every $a = |A|, b = |B|, c = |C|, d = |D| > 0$, every real $1 < \alpha \leq 2$, and every ordered pair of physical deterministic superchannels as in (5),*

$$\mathcal{A}_\alpha(\Theta_1 \parallel \Theta_2) = U_\alpha(\Theta_1 \parallel \Theta_2). \quad (8)$$

Equality is in the extended real numbers and has no additional support, invertibility, common-memory, or channel-collapse hypothesis.

Proposition 2 (Ordinary reference absorption). *The ordinary quantity also has the output-channel form*

$$U_\alpha(\Theta_1 \parallel \Theta_2) = \sup_{R, N: AR \rightarrow BR} d_\alpha(\Theta_1^R(N) \parallel \Theta_2^R(N)). \quad (9)$$

Proof. An ordinary state test in (6) is a channel test with a one-dimensional S , giving one inequality. Conversely, fix a channel test with references R, S and state ρ_{SCR} . Set $R' = R \otimes S$ and, after the canonical tensor permutations, insert $N' = N \otimes \text{id}_S : AR' \rightarrow BR'$. Tensoring each tooth of (5) by id_S shows, on every matrix,

$$\Theta_i^{R'}(N') = \Theta_i^R(N) \otimes \text{id}_S$$

in the reordered bases, simultaneously for $i = 1, 2$. Move the state to CR' using the same permutation. The output pair is the channel-test pair up to a unitary permutation, preserving both supports and (2). This supplies an ordinary test for every external-reference channel test and proves the reverse inequality. This argument is independent of amortization collapse; it is a mathematical correspondence argument, not a separately registered Lean theorem in this artifact. $\square$

3 The supported matrix transformer estimate

Lemma 3 (Transformer and congruence identities). *If $X, Y \geq 0$ and $X \ll Y$, then for every rectangular matrix K ,*

$$KXK^\dagger \ll KYK^\dagger, \quad G_\alpha(KXK^\dagger, KYK^\dagger) \leq KG_\alpha(X, Y)K^\dagger. \quad (10)$$

If K has a two-sided inverse, the operator inequality is an equality and support inclusion is equivalent before and after congruence. Moreover,

$$G_\alpha(X_1 \otimes X_2, Y_1 \otimes Y_2) = G_\alpha(X_1, Y_1) \otimes G_\alpha(X_2, Y_2), \quad (11)$$

$$G_\alpha(X, X) = X. \quad (12)$$

The tensor identity holds for positive semidefinite arguments, and the corresponding tensor pair is supported whenever both factors are.

Proof. First consider $\alpha = 2$. On a supported pair, $G_2(X, Y) = XY^+X$. The block matrix

$$\begin{pmatrix} Y & X \\ X & XY^+X \end{pmatrix} = \begin{pmatrix} Y^{1/2} \\ XY^{+1/2} \end{pmatrix} \begin{pmatrix} Y^{1/2} \\ XY^{+1/2} \end{pmatrix}^\dagger \geq 0. \quad (13)$$

Here the support relation makes the off-diagonal blocks equal to X . In a positive block matrix $\begin{pmatrix} Y' & X' \\ X' & T' \end{pmatrix}$, a vector in $\ker Y'$ gives a zero quadratic form on the upper block and therefore belongs to $\ker X'$. Thus $X' \ll Y'$. Evaluating the block quadratic form on $(-Y'^+X'v, v)$ gives $T' \geq X'Y'^+X'$. Apply these facts to the congruence of (13) by $\text{diag}(K, K)$ to obtain (10) at order two, including singular denominators.

For $1 < \alpha < 2$, use the positive scalar representation

$$x^\alpha = c_\alpha \int_0^\infty t^{\alpha-2} \frac{x^2}{x+t} dt, \quad c_\alpha = \frac{\sin(\pi(\alpha-1))}{\pi} > 0, \quad x \geq 0. \quad (14)$$

For $x > 0$ substitution $t = xu$ and the beta integral give this identity; for $x = 0$ both sides vanish. The integrand is $O(t^{\alpha-2})$ near zero and $O(t^{\alpha-3})$ at infinity, hence is integrable in the stated range. Diagonalize $Z = Y^{+1/2}XY^{+1/2}$. On $\text{supp } Y$, Y is invertible and

$$X(X + tY)^+X = Y^{1/2}Z(Z + tI)^{-1}ZY^{1/2}, \quad t > 0.$$

Both sides vanish on the complementary kernel. Applying (14) to each eigenvalue of Z therefore gives the integrable quadratic-form identity

$$v^\dagger G_\alpha(X, Y)v = c_\alpha \int_0^\infty t^{\alpha-2} v^\dagger X(X + tY)^+Xv dt. \quad (15)$$

This is a finite sum of the integrable scalar expressions just considered.

For every $t > 0$, $X \ll X + tY$. Apply the order-two transformer estimate to $(X, X + tY)$; its congruent denominator is $KXK^\dagger + tKYK^\dagger$. Integrating its quadratic forms against the positive weight in (15) proves (10). Support preservation already follows from the block argument. For invertible K , applying the inequality again with K^{-1} gives the reverse operator inequality; the same reverse congruence proves support equivalence.

Finally, spectral calculus on tensor-product eigenbases gives $(Y_1 \otimes Y_2)^{+1/2} = Y_1^{+1/2} \otimes Y_2^{+1/2}$ and $(Z_1 \otimes Z_2)^\alpha = Z_1^\alpha \otimes Z_2^\alpha$, proving (11). For equal arguments the middle sandwich is the support projection of X , whose α th power is itself, proving (12). Tensor support inclusion follows, for example, by writing $X_i = Y_i Z_i'$ and tensoring these factorizations. $\square$

This transformer statement is the established estimate of Fang–Fawzi [2, Lemma 47]; the preceding proof follows the Lean construction through block positivity and the all-real-order integral. Lean uses Mathlib’s positive representing measure for the scalar power identity, rather than fixing the explicit density $c_\alpha dt$.

Lemma 4 (Normalization and data processing). *Supported states satisfy $M_\alpha(\rho, \sigma) \geq 1$ and $\widehat{D}_\alpha(\rho||\rho) = 0$. For every channel N ,*

$$\widehat{D}_\alpha((\text{id}_S \otimes N)(\rho)||(\text{id}_S \otimes N)(\sigma)) \leq \widehat{D}_\alpha(\rho||\sigma), \quad d_\alpha^A(N||N) = 0. \quad (16)$$

Proof. Diagonalize $Z = \sigma^{+1/2} \rho \sigma^{+1/2}$. The diagonal entries of σ in this eigenbasis are nonnegative weights summing to one, and their weighted average of the eigenvalues of Z is $\text{Tr } \rho = 1$. Scalar convexity of x^α gives $M_\alpha(\rho, \sigma) \geq 1$. Equation (12) gives the assertion on equal states.

Use the input-first unnormalized channel Choi matrix

$$(J_N)_{(i,\beta),(j,\gamma)} = N(|i\rangle\langle j|)_{\beta\gamma}, \quad J_N \geq 0, \quad \text{Tr}_Q J_N = I_P. \quad (17)$$

The amplified action on Z' has entries

$$[\mathcal{C}(J_N, Z')]_{(\ell,\beta),(\ell',\gamma)} = \sum_{i,j} (J_N)_{(i,\beta),(j,\gamma)} Z'_{(\ell,i),(\ell',j)}.$$

It is a congruence of $J_N \otimes Z'$, obtained by linking the repeated input labels. Apply Lemma 3, (11), and (12) to $(J_N \otimes \rho, J_N \otimes \sigma)$. The output pair is supported, and its moment is at most $\text{Tr } \mathcal{C}(J_N, G_\alpha(\rho, \sigma)) = \text{Tr } G_\alpha(\rho, \sigma)$, using $\text{Tr}_Q J_N = I_P$. Taking logarithms proves (16) on finite-cost pairs; infinite input divergence makes the inequality automatic. Every admissible term in $d_\alpha^A(N||N)$ is therefore at most zero, while equal input states supply the value zero. This proves its asserted value. $\square$

4 The full finite input-channel cost

Lemma 5 (Choi moment bound from the entire cost). *Let $N, M : P \rightarrow Q$, with $p = |P| > 0$, and suppose $d_\alpha^A(N||M) = t \in \mathbb{R}$. Then*

$$J_N \ll J_M, \quad \text{Tr}_Q G_\alpha(J_N, J_M) \leq 2^{\kappa t} I_P. \quad (18)$$

Proof. For every invertible K on P with $\text{Tr } K^\dagger K = 1$, use the equal pure input states $\phi_K = |\phi_K\rangle\langle\phi_K|$, where $|\phi_K\rangle = (K \otimes I_P) \sum_i |i\rangle|i\rangle$. Their input divergence is zero, so their output divergence is at most the full supremum t . The two output states are $(K \otimes I_Q) J_N (K^\dagger \otimes I_Q)$ and the corresponding expression for J_M . The choice $K = I_P/\sqrt{p}$ and finiteness of t force $J_N \ll J_M$ by invertible congruence.

Set $F = G_\alpha(J_N, J_M)$ and $E = \text{Tr}_Q F$. Congruence equality from Lemma 3 gives, for every such K ,

$$\text{Tr}(EK^\dagger K) = M_\alpha((\text{id} \otimes N)(\phi_K), (\text{id} \otimes M)(\phi_K)) \leq 2^{\kappa t}.$$

Every positive definite trace-one η has the form $K^\dagger K$ for an invertible K . For a unit vector v , apply this trace bound to $\eta_\varepsilon = (1 - \varepsilon)|v\rangle\langle v| + \varepsilon I_P/p$ and let $\varepsilon \downarrow 0$. It follows that $v^\dagger E v \leq 2^{\kappa t}$ for every v , which is exactly (18). No channel amortization-collapse assumption was used. $\square$

5 Comb contractions and tester completion

Fix computational bases and order the comb space as $C \otimes A \otimes B \otimes D$. Define J_Θ from the two teeth of (5) by

$$(J_\Theta)_{(j,i,k,\delta),(j',i',k',\delta')} = \sum_{e,f} (J_{P_\Theta})_{(j,i,e),(j',i',f)} (J_{Q_\Theta})_{(k,e,\delta),(k',f,\delta')}. \quad (19)$$

This is positive semidefinite: using Kraus operators p_h, q_g of the two teeth, it is $\sum_{h,g} |v_{hg}\rangle\langle v_{hg}|$ with amplitudes $v_{hg}(j, i, k, \delta) = \sum_e (p_h)_{ie,j} (q_g)_{\delta,ke}$.

Write $x = (j, i, k) \in C \otimes A \otimes B$. Given an insertion reference R and state reference S , define the trilinear contraction

$$\begin{aligned} [\mathcal{L}_{r,s}(J, F, H)]_{(\ell, \delta, t), (\ell', \delta', t')} & \\ = \sum_{x, y, u, v} J_{(x, \delta), (y, \delta')} F_{(i, u; k, t), (i', v; k', t')} H_{(\ell, j, u), (\ell', j', v)}, \end{aligned} \quad (20)$$

where $y = (j', i', k')$, u, v, t, t' are reference labels in R , and ℓ, ℓ' are labels in S . Expanding the two teeth and the inserted channel on matrix units gives the full action identity

$$(\text{id}_S \otimes \Theta^R(N))(\rho) = \mathcal{L}_{r,s}(J_\Theta, J_N, \rho). \quad (21)$$

It holds for every matrix and every complex-linear insertion. In particular it is not limited to product states or product insertions.

For completeness, (20) is a congruence $K_{r,s}(J \otimes F \otimes H)K_{r,s}^\dagger$, where

$$K_{r,s} = \sum_{\ell, \delta, t} \sum_{j, i, k, u} |\ell, \delta, t\rangle \langle j, i, k, \delta| \otimes \langle i, u; k, t| \otimes \langle \ell, j, u|.$$

Thus, for three supported positive pairs, Lemma 3 gives

$$\begin{aligned} M_\alpha(\mathcal{L}(J_1, J_N, \rho), \mathcal{L}(J_2, J_M, \sigma)) \\ \leq \text{Tr } \mathcal{L}(G_\alpha(J_1, J_2), G_\alpha(J_N, J_M), G_\alpha(\rho, \sigma)), \end{aligned} \quad (22)$$

and also gives support inclusion for the contracted pair.

For W, Γ on $C \otimes A \otimes B$ use the full-transpose pairing

$$\langle W, \Gamma \rangle_{\text{T}} = \Re \text{Tr}(W \Gamma^{\text{T}}) = \Re \sum_{x, y} W_{xy} \Gamma_{xy}. \quad (23)$$

This uses the transpose of the entire tester in the fixed basis, not a partial transpose. If $W, \Gamma \geq 0$ it is nonnegative; if $\Gamma' \geq \Gamma$ it is monotone. Call Γ feasible if

$$\Gamma \geq 0, \quad \text{Tr}_B \Gamma = \tau_C \otimes I_A, \quad \tau_C \geq 0, \quad \text{Tr } \tau_C = 1. \quad (24)$$

Lemma 6 (Tester completion). *Let $J, F, H \geq 0$, put $W = \text{Tr}_D J$, and suppose*

$$\text{Tr}_{BR} F \leq q I_{AR}, \quad q > 0, \quad \text{Tr } H = m > 0.$$

If $\langle W, \Gamma \rangle_{\text{T}} \leq Q$ for every feasible tester, then

$$\text{Tr } \mathcal{L}_{r,s}(J, F, H) \leq qmQ. \quad (25)$$

Proof. Define the contracted tester T on $C \otimes A \otimes B$ by

$$T_{(j, i, k), (j', i', k')} = \sum_{t, \ell, u, v} F_{(i, u; k, t), (i', v; k', t)} H_{(\ell, j, u), (\ell, j', v)}. \quad (26)$$

For each t, ℓ this is a congruence of $F \otimes H$, linking the reference label, and summing these congruences proves $T \geq 0$. Taking the trace in (20) yields

$$\text{Tr } \mathcal{L}_{r,s}(J, F, H) = \langle W, T \rangle_{\text{T}}. \quad (27)$$

Both traces here are real for the positive arguments.

Let $Y = \text{Tr}_{SR} H$, a positive operator on C with $\text{Tr} Y = m$. The B marginal of T is obtained by the same positive contraction from $\text{Tr}_{BR} F$ and H . More explicitly, the relevant map sends E on AR to the matrix on CA with entries

$$\mathcal{B}_H(E)_{(j,i),(j',i')} = \sum_{\ell,u,v} E_{(i,u),(i',v)} H_{(\ell,j,u),(\ell,j',v)}.$$

For $E \geq 0$, its value is a sum of congruences of $E \otimes H$, hence is positive. It satisfies $\mathcal{B}_H(\text{Tr}_{BR} F) = \text{Tr}_B T$ and $\mathcal{B}_H(I_{AR}) = Y \otimes I_A$. Consequently

$$\Delta = q(Y \otimes I_A) - \text{Tr}_B T \geq 0.$$

Complete and normalize the tester by

$$\tilde{\Gamma} = \frac{T + \Delta \otimes I_B/b}{qm}. \quad (28)$$

It is positive, its B marginal is $(Y/m) \otimes I_A$, and Y/m is a state, so it is feasible. Since the added slack is positive, $\langle W, T \rangle_T \leq qm \langle W, \tilde{\Gamma} \rangle_T \leq qmQ$. Together with (27) this proves the bound for all r, s . $\square$

Proposition 7 (Uniform bound for nested amortization). *Suppose $J_1 = J_{\Theta_1} \ll J_2 = J_{\Theta_2}$ and set $W_\alpha = \text{Tr}_D G_\alpha(J_1, J_2)$. If $Q > 0$ bounds $\langle W_\alpha, \Gamma \rangle_T$ for every feasible tester, then*

$$\mathcal{A}_\alpha(\Theta_1 \| \Theta_2) \leq \kappa^{-1} \log_2 Q. \quad (29)$$

Proof. Fix an admissible pair $N, M : AR \rightarrow BR$ and write its entire cost as $t = d_\alpha^A(N \| M) \in \mathbb{R}$. By Lemma 5, $J_N \ll J_M$ and $F = G_\alpha(J_N, J_M)$ has output marginal at most qI_{AR} , with $q = 2^{\kappa t}$. Fix any admissible inner test ρ, σ on SCR . Its finite cost gives $\rho \ll \sigma$, and $H = G_\alpha(\rho, \sigma)$ has trace $m = M_\alpha(\rho, \sigma) \geq 1$. Equations (21) and (22), followed by Lemma 6, give output support inclusion and

$$M_\alpha((\text{id}_S \otimes \Theta_1^R(N))(\rho), (\text{id}_S \otimes \Theta_2^R(M))(\sigma)) \leq 2^{\kappa t} M_\alpha(\rho, \sigma) Q. \quad (30)$$

The supported output states have moment at least one. Taking logarithms therefore yields

$$\hat{D}_\alpha(\text{output}_1 \| \text{output}_2) - \hat{D}_\alpha(\rho \| \sigma) \leq t + \kappa^{-1} \log_2 Q.$$

Take the inner supremum, subtract exactly t , then take the outer supremum. This proves (29). All support and reference conditions used above were obtained from the admissible finite costs. $\square$

6 Physical realization of testers

Lemma 8 (Common-reference realization). *For every positive definite feasible tester Γ whose marginal state τ is positive definite, there are a reference R of dimension $r_* = \text{cab}$, a channel $N : AR \rightarrow BR$, a pure state ρ_{CR} , and an invertible matrix $K_T : CABD \rightarrow DR$ such that, for every physical Θ ,*

$$\Theta^R(N)(\rho) = K_T J_\Theta K_T^\dagger. \quad (31)$$

For supported $J_1 \ll J_2$, this gives

$$M_\alpha(\Theta_1^R(N)(\rho), \Theta_2^R(N)(\rho)) = \langle W_\alpha, \Gamma \rangle_T \geq 1. \quad (32)$$

For an unsupported comb pair, the same test has unsupported outputs.

Proof. Choose invertible factors T, ψ with $\Gamma = TT^\dagger$ and $\tau = \psi\psi^\dagger$. Identify the basis of R with $C \otimes A \otimes B$, and fix labels i_0, k_0 to embed $\iota : C \hookrightarrow R$, $u \mapsto (u, i_0, k_0)$. Prepare

$$|\phi\rangle = \sum_{j,u} \psi_{ju} |j\rangle_C |\iota(u)\rangle_R, \quad \rho = |\phi\rangle\langle\phi|. \quad (33)$$

Its norm is one because $\text{Tr } \tau = 1$.

On the input subspace $A \otimes \iota(C)$, define a matrix $V : AR \rightarrow BR$ by

$$V_{(k,t),(i,\iota(u))} = \sum_j (\psi^{-1})_{uj} T_{(j,i,k),t}, \quad (34)$$

and set it to zero on the orthogonal complement. The marginal equation for $TT^\dagger$ gives

$$\begin{aligned} \sum_{k,t} V_{(k,t),(i,\iota(u))} \overline{V_{(k,t),(i',\iota(u'))}} &= \delta_{ii'} [\psi^{-1} \tau (\psi^{-1})^\dagger]_{uu'} \\ &= \delta_{ii'} \delta_{uu'}. \end{aligned}$$

Thus V is an isometry on that subspace. If P is its input projection and $|\beta_0\rangle$ a fixed unit vector in BR , set

$$N(X) = V X V^\dagger + \text{Tr}((I - P)X) |\beta_0\rangle\langle\beta_0|. \quad (35)$$

The reset term has Kraus operators $|\beta_0\rangle\langle i, w|$ for basis vectors outside the embedded subspace. Their adjoint products sum to $I - P$, while $V^\dagger V = P$. Hence N is completely positive and trace preserving on the entire input space, including unused input vectors. The reset term vanishes on the chosen initial state's reference support and after the preprocessing tooth.

The decisive contraction identity is

$$\sum_u V_{(k,t),(i,\iota(u))} \psi_{ju} = T_{(j,i,k),t}.$$

Insert it and its conjugate into (20), with S trivial. This proves (31), where

$$(K_T)_{(\delta,t),(x,\delta')} = \delta_{\delta\delta'} T_{x,t}. \quad (36)$$

Since $r_* = cab$, this is, up to a tensor permutation, $T^\top$ tensored with I_D . It has a two-sided inverse obtained from $(T^{-1})^\top$ and the reverse permutation.

For supported J_1, J_2 , Lemma 3 gives perspective equality under this congruence. Taking traces and summing the output reference label,

$$\begin{aligned} \text{Tr } K_T G_\alpha(J_1, J_2) K_T^\dagger &= \sum_{x,y,\delta,t} G_\alpha(J_1, J_2)_{(x,\delta),(y,\delta)} T_{x,t} \overline{T_{y,t}} \\ &= \langle W_\alpha, TT^\dagger \rangle_T = \langle W_\alpha, \Gamma \rangle_T. \end{aligned}$$

Both outputs are normalized physical states, giving the last inequality in (32). For an unsupported pair, invertibility of K_T preserves failure of support inclusion, so its output divergence is $+\infty$. The construction depends on the tester, not on a choice of superchannel realization, and works simultaneously for both members of the pair with their separate memories. $\square$

7 Assembly of the equality and the tester formula

Proof of Theorem 1. Lemma 4 shows that equal insertions $N = M$ have entire input cost zero. For each ordinary state test choose these equal insertions in the outer amortization and equal states with a trivial external reference in the inner amortization. Consequently, without any comb support assumption,

$$U_\alpha(\Theta_1 \parallel \Theta_2) \leq \mathcal{A}_\alpha(\Theta_1 \parallel \Theta_2). \quad (37)$$

First assume $J_1 \ll J_2$ and use W_α as above. Every feasible tester has trace a , so positivity gives

$$0 \leq \langle W_\alpha, \Gamma \rangle_T \leq a \operatorname{Tr} W_\alpha. \quad (38)$$

Indeed, for positive B, P , $\operatorname{Tr} BP \leq \operatorname{Tr} B \operatorname{Tr} P$, which follows by diagonalizing P and bounding each eigenvalue by its trace. Apply this with $P = \Gamma^\top$.

There is a positive definite feasible tester

$$\Gamma_* = (I_C/c) \otimes I_A \otimes (I_B/b) = I_{CAB}/(cb). \quad (39)$$

Its realization has moment $\langle W_\alpha, \Gamma_* \rangle_T \geq 1$. Thus $Q_0 = a \operatorname{Tr} W_\alpha$ is positive and finite. Equations (29) and (37) bound both quantities above by $\kappa^{-1} \log_2 Q_0$.

Let U_{r_*} denote the ordinary supremum with the inserted reference fixed to $r_* = cab$. The realization of Γ_* supplies a finite lower bound for this supremum, and $U_{r_*} \leq U_\alpha \leq \mathcal{A}_\alpha \leq \kappa^{-1} \log_2 Q_0$. Hence $U_{r_*} = y$ for some finite real number y . Realization of every positive definite feasible tester gives

$$\kappa^{-1} \log_2 \langle W_\alpha, \Gamma \rangle_T \leq y, \quad \langle W_\alpha, \Gamma \rangle_T \leq 2^{\kappa y}.$$

To extend this bound to any feasible Γ , form $\Gamma_\varepsilon = (1 - \varepsilon)\Gamma + \varepsilon\Gamma_*$ with $0 < \varepsilon < 1$. It is positive definite and feasible, with positive definite marginal $(1 - \varepsilon)\tau + \varepsilon I_C/c$. The pairing is affine in ε , so letting $\varepsilon \downarrow 0$ gives the same bound for Γ . This step uses only an affine scalar expression; no continuity of inverse powers at a singular denominator is assumed.

Apply Proposition 7 with $Q = 2^{\kappa y}$ to conclude $\mathcal{A}_\alpha \leq y$. Therefore

$$y = U_{r_*} \leq U_\alpha \leq \mathcal{A}_\alpha \leq y,$$

proving collapse and the reference-dimension assertion in the supported case.

If $J_1 \not\ll J_2$, realize Γ_* instead. Its invertible congruence gives unsupported ordinary output states, so $U_\alpha = +\infty$. Equation (37) then gives $\mathcal{A}_\alpha = +\infty$ as well. The lower-bound test has zero finite input cost, so no subtraction of infinite values occurs. This proves the theorem for all physical pairs. $\square$

Corollary 9 (Finite tester characterization). *For $J_1 \ll J_2$ put*

$$Q_\alpha = \sup_{\Gamma \text{ feasible}} \langle \operatorname{Tr}_D G_\alpha(J_1, J_2), \Gamma \rangle_T. \quad (40)$$

Then $1 \leq Q_\alpha < \infty$ and

$$\mathcal{A}_\alpha = U_\alpha = U_{cab} = \frac{1}{\alpha - 1} \log_2 Q_\alpha. \quad (41)$$

For unsupported comb pairs both divergences are $+\infty$, already witnessed at reference dimension cab . The asserted reference bound concerns an equality of suprema; it makes no claim that an optimizing physical test is attained.

Proof. The proof of the theorem gives a finite y with all tester pairings at most $2^{\kappa y}$. Any other upper bound Q' is at least the pairing of Γ_* , hence $Q' \geq 1$. Proposition 7 then gives $y = \mathcal{A}_\alpha \leq \kappa^{-1} \log_2 Q'$, so $2^{\kappa y} \leq Q'$. Thus $2^{\kappa y}$ is the least upper bound in (40). The unsupported assertion follows from the same realization used above. $\square$

8 Formal verification and correspondence

The frozen artifact contains the proved theorem declaration

```
OpenQ.Problems.
AmortizationCollapseSuperchannelDivergences_148275.
referenceStabilized_collapse
```

with exact theorem type `ReferenceStabilizedMainStatement`. This named proposition unfolds to the expanded theorem statement, a quantified equality of the two outer optimization domains. It quantifies all positive finite a, b, c, d , every real $1 < \alpha \leq 2$, and every pair of bundled physical realizations. It imposes no extra premise. Its state divergence uses the actual matrix-range support relation, real spectral powers, the support inverse square root, and base-two logarithms. Each finite subtracted cost is recorded by a real witness equal to the entire corresponding supremum, exactly as in (4) and (7).

The conventional proof follows the formal proof’s sequence of supported transformer bounds, finite input-cost control, tester completion, physical realization, and assembly through the ordinary value at dimension cab . The following dictionary uses names relative to the namespace displayed above. It distinguishes the standard reference-absorption argument of Proposition 2, which is supplied here as a mathematical correspondence argument, from the separately formalized collapse.

Mathematical step	Lean module and declarations
Definitions and theorem domain	<code>Statement</code> : <code>PhysicalSuperchannel</code> , <code>geometricStateDivergence</code> , <code>amortizedChannelDivergence</code> ; <code>StabilizedStatement</code> : <code>referenceStabilizedOrdinaryDivergence</code> , <code>referenceStabilizedAmortizedDivergence</code> , <code>ReferenceStabilizedMainStatement</code> .
Reference action	<code>StabilizedAction</code> : <code>PhysicalSuperchannel.referenceRealization</code> , <code>referenceAct_formula</code> , <code>referenceAct_isCPTP</code> .
Lemma 3	<code>StabilizedProofDom</code> : <code>dom_base</code> , <code>dom_range</code> , <code>dom_schur</code> ; <code>StabilizedProofChannelValue</code> : <code>transformer_le</code> ; <code>StabilizedProofAlpha</code> : <code>qform_integral</code> , <code>transformer_alpha</code> , <code>Gp_conj_eq</code> ; <code>StabilizedProofAlphaChain</code> : <code>Gp_kron</code> , <code>Gp_self</code> .
Lemma 4	<code>ProofNonnegative</code> : <code>one_le_geometricMoment</code> ; <code>StabilizedProofAlphaMain</code> : <code>data_processing_alpha</code> , <code>amortized_self_alpha</code> .
Lemma 5	<code>StabilizedProofAlphaMain</code> : <code>psd_of_trace_tests</code> , <code>cost_bound_alpha</code> .
Comb and full action	<code>StabilizedProofComb</code> : <code>combJ</code> , <code>combJ_psd</code> , <code>refAct_entry</code> , <code>amp_refAct</code> ; <code>StabilizedProofAlphaChain</code> : <code>chainS_alpha</code> .
Lemma 6	<code>StabilizedProofTester</code> : <code>Feasible</code> , <code>tester_psd</code> , <code>trace_contrS</code> , <code>ptr_tester</code> , <code>chain_trace_bound</code> .
Proposition 7	<code>StabilizedProofAlphaMain</code> : <code>moment_chain_alpha</code> , <code>amortized_le_alpha</code> .
Lemma 8	<code>StabilizedProofRealize</code> : <code>Data.link</code> , <code>Data.rho_density</code> , <code>Data.JN_psd</code> , <code>Data.JN_marg</code> , <code>Data.output</code> , <code>Data.KT'_mul_KT</code> , <code>Data.KT_mul_KT'</code> ; <code>StabilizedProofOrderTwo</code> : <code>exists_data</code> ; <code>StabilizedProofAlphaMain</code> : <code>realized_alpha</code> .
Singular tester extension	<code>StabilizedProofOrderTwo</code> : <code>gammaStar_feasiblePD</code> , <code>pairing_le_of_PD</code> .
Theorem and Corollary	<code>StabilizedProofAlphaMain</code> : <code>ordinary_le_amortized_alpha</code> , <code>regular_alpha</code> , <code>collapse_alpha</code> , <code>referenceStabilizedMainStatement_holds</code> ; <code>StabilizedCollapse</code> : <code>referenceStabilized_collapse</code> .

The verification script rebuilds all 43 project modules in the import closure from the frozen sources into an initially empty directory. Its `LEAN_PATH` excludes existing project and critic proof

caches. It checks that the declaration is a theorem of exactly the named type, with no additional theorem-level or universe parameters. It separately checks an independently written expansion of the outer optimization domains and enforces an axiom allowlist on the reports for the main theorem and five critical exports. The reported axiom set is `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx` occurs. It then replays the newly built closure using `leanchecker`, and verifies that the source hashes remain unchanged. The publication run passed these checks on the frozen source copy.

The checked toolchain is Lean 4.34.1. Mathlib and Physlib are pinned to, respectively, revisions

```
d13f23b723b8a846827a245b89c10fc7d3f11612
44c66d54be78db4693be9f8f92bd3b5ad124ed6f.
```

Installed Lean and external package binaries were retained; the publication run checked their dependency source revisions and clean tracked sources but did not rebuild or replay the external packages. Kernel replay uses the installed Lean kernel, not a separately implemented verifier. These are the verification trust boundaries. Formal checking proves the encoded theorem; its correspondence with the physical question is established by the definitions and mathematical arguments above.

9 Contributions, AI disclosure, and availability

Joint AI research and human role. The result and its proofs were developed through joint work by GPT-6.1-Sol and Claude Opus 5.5 agents. Vinícius Mohr’s role in the discovery was to direct the AI agents to attempt open problems listed on the Quantum Information and Quantum Computation Open Problem Zoo (<https://qiqc-op.com/>) using the `lean-orchestrator` repository (<https://github.com/vimohr/lean-orchestrator>), which was itself built using Claude Opus 5.5. The role assignments in `agents.toml` use GPT-6.1-Sol for the researcher, supervisor, literature, and triage agents, and Claude Opus 5.5 for the critic agent. The agents carried out mathematical exploration, Lean formalization, proof development, critical review, and literature searches. The critic also produced substantive proof material, as recorded below. Mohr subsequently requested verification and manuscript preparation.

Manuscript preparation. GPT-6.1-Sol wrote this document, including the conventional proof exposition reconstructed from the formal sources.

The registered theorem incorporates 16 modules ported from earlier local critic-agent proof material produced during iterations `e002-i01` and `e002-i02`. The registration wrapper applies that assembled proof; it does not constitute a second independently discovered proof. A later audit and the publication build checked its sources and interpretation independently of the original agent’s acceptance verdict. The conventional proof in this manuscript was reconstructed from those sources. Mathematical ingredients from Fang–Fawzi and the physical realization theorem are credited separately above.

License. This manuscript and the accompanying project source code and associated documentation are licensed under the MIT License. The source release contains the full license text in `LICENSE`. External dependencies retain their respective licenses.

Availability. The manuscript’s Zenodo DOI is <https://doi.org/10.5281/zenodo.23215121>. The manuscript sources, complete project proof closure, verification script, and recorded logs are publicly available at <https://github.com/vimohr/superchannel-amortization>. The repository’s `VERIFY.md` and `HUMAN-CHECK.md` explain how to reproduce the checks and inspect the definitions. Its source manifest records every project module’s SHA-256 hash. The formal verification

artifact is permanently archived on Zenodo; the specific version corresponding to release v1.0.1 is available at <https://doi.org/10.5281/zenodo.23262928>.

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